CS5234 Mock Quiz 2

Lectures 6–11 · k-means not examined · 50 minutes · 20 points 第 6–11 讲 · k-means 不考 · 50 分钟 · 共 20 分

Questions 1–4: one correct option each (1 point). Questions 5–6: show your reasoning; an essay question left completely blank receives 1 mark. log denotes the natural log unless stated.

第 1–4 题单选，每题 1 分。第 5–6 题需写出推理；完全空白的大题给 1 分。log 默认为自然对数。

1 [1 pt]

A stream of updates (i, Δ) with arbitrary integer Δ (frequencies may become negative). You want, for every i, an estimate with $|\tilde{x}_i - x_i| \le \varepsilon \|x\|_2$. Which sketch should you use?

一串更新 (i, Δ)，Δ 为任意整数（频率可以变成负数）。要对每个 i 给出估计，满足 $|\tilde{x}_i - x_i| \le \varepsilon \|x\|_2$。应选哪个 sketch?

(A) Misra-Gries

(B) Count-Min

(C) Count Sketch

(D) KMV

2 [1 pt]

In the KMV distinct-elements algorithm (output tm/X where X is the t-th smallest hash value), what is the weakest independence of the hash function that the lecture's analysis needs?

KMV 不同元素算法（输出 tm/X，X 为第 t 小的哈希值）中，课上的分析最少需要哈希函数满足哪种独立性?

(A) Fully (mutually) independent 完全独立

(B) 4-wise independent 4-wise 独立

(C) Pairwise independent 两两独立

(D) No independence is needed 不需要任何独立性

3 [1 pt]

Which statement about metric k-center is true (assuming P $\neq$ NP)?

关于度量空间上的 k-center，哪项正确（假设 P $\neq$ NP）？

(A) Farthest-first greedy always finds an optimal solution. 最远点贪心总能找到最优解。

(B) Farthest-first greedy is a 2-approximation, and no polynomial-time algorithm achieves ratio 1.9. 最远点贪心是 2-近似，且没有多项式算法能达到 1.9 的近似比。

(C) For every $\varepsilon > 0$ there is a polynomial-time $(1+\varepsilon)$-approximation. 对任意 $\varepsilon > 0$ 都有多项式时间的 $(1+\varepsilon)$-近似。

(D) The approximation ratio of farthest-first greedy grows with k. 最远点贪心的近似比随 k 增大而变差。

4 [1 pt]

A connected graph has $n = 10$ vertices and integer edge weights in $\{1, 2, 3\}$. Let G_i be the subgraph with edges of weight $\le i$. You are told $\#CC(G_1) = 4$ and $\#CC(G_2) = 2$. What is the weight of the MST?

连通图有 $n = 10$ 个点，边权为 $\{1, 2, 3\}$ 中的整数。G_i 为权 $\le i$ 的边构成的子图。已知 $\#CC(G_1) = 4$，$\#CC(G_2) = 2$。MST 的权重是多少?

(A) 11

(B) 12

(C) 13

(D) 16

5 [9 pts]

Two vectors $x, y \in \mathbb{Z}^m$ start at 0. A single stream interleaves updates (x, i, Δ) meaning $x_i \mathrel{+}= \Delta$ and (y, i, Δ) meaning $y_i \mathrel{+}= \Delta$, with arbitrary integer Δ (turnstile). All entries stay bounded by poly(n). Design a streaming algorithm that outputs $D = \|x - y\|_2^2$ within $(1 \pm \varepsilon)$ with probability $1 - \delta$.

两个向量 $x, y \in \mathbb{Z}^m$ 初始为 0。同一条流里交替出现更新 (x, i, Δ) 表示 $x_i \mathrel{+}= \Delta$，(y, i, Δ) 表示 $y_i \mathrel{+}= \Delta$，Δ 为任意整数（turnstile）。所有分量绝对值不超过 poly(n)。设计流算法，以概率 $1 - \delta$ 输出 $D = \|x - y\|_2^2$ 的 $(1 \pm \varepsilon)$ 近似。

(a, 2 pts) Describe one basic estimator: what is stored and how each update is processed. Why must x and y use the *same* random signs?

描述一个基本估计量：存什么、每条更新怎么处理。为什么 x 和 y 必须用同一组随机符号？

(b, 4 pts) Show $\mathbb{E}[Z^2] = D$ and $\mathrm{Var}[Z^2] \le 2D^2$. State exactly where you use 4-wise independence.

证明 $\mathbb{E}[Z^2] = D$ 且 $\mathrm{Var}[Z^2] \le 2D^2$，并指出在哪里用到 4-wise 独立。

(c, 3 pts) Choose the number of copies and give the total space in bits.

确定副本数，给出总空间（位）。

6 [7 pts]

In problem DUP, the stream is a sequence of numbers from $[m]$ and the algorithm must say whether some number appears at least twice. You may use: any randomized one-way protocol for Index on m bits with error $\le 1/3$ needs $\Omega(m)$ bits.

问题 DUP：流是一串 $[m]$ 中的数，算法要判断是否有某个数出现至少两次。可直接使用：m 位 Index 问题的任何错误率 $\le 1/3$ 的随机单向协议都需要 $\Omega(m)$ 位。

(a, 4 pts) Show that every one-pass streaming algorithm for DUP that succeeds with probability $\ge 2/3$ uses $\Omega(m)$ bits of memory.

证明：任何成功率 $\ge 2/3$ 的单遍 DUP 流算法都需要 $\Omega(m)$ 位内存。

(b, 1 pt) Why is it fine that the streaming algorithm is randomized?

为什么流算法是随机的也没关系？

(c, 2 pts) Use the same idea to show that computing the *exact* number of distinct elements DE needs $\Omega(m)$ bits.

用同样的思路证明：精确计算不同元素个数 DE 需要 $\Omega(m)$ 位。